

Extracted Information for 2D Ising CFT 5-Point Function

Based on the standard literature for the two-dimensional Ising Conformal Field Theory (CFT), specifically Di Francesco et al. (1997) [1], the necessary theoretical information to compute the 5-point correlation function $\langle \epsilon(x_1)\epsilon(x_2)\epsilon(x_3)\sigma(x_4)\sigma(x_5) \rangle$ is as follows:

1. Operator Content and Conformal Dimensions

The model is a minimal model $\mathcal{M}(3, 4)$ with central charge $c = 1/2$. The primary fields involved in the correlation function are:

- **Spin Field (σ):**
 - Conformal weight: $\Delta_\sigma = \bar{\Delta}_\sigma = \frac{1}{16}$
 - Scaling dimension: $\Delta + \bar{\Delta} = \frac{1}{8}$
- **Energy Field (ϵ):**
 - Conformal weight: $\Delta_\epsilon = \bar{\Delta}_\epsilon = \frac{1}{2}$
 - Scaling dimension: $\Delta + \bar{\Delta} = 1$
- **Identity Field ($\mathbf{1}$):**
 - Conformal weight: $\Delta_1 = 0$

2. Fusion Rules (Operator Product Expansion)

The structure of the correlation function is determined by the fusion rules (OPE channels) allowed by the theory:

- $\sigma \times \sigma = \mathbf{1} + \epsilon$
- $\sigma \times \epsilon = \sigma$
- $\epsilon \times \epsilon = \mathbf{1}$

These rules imply that the 5-point function can be decomposed into conformal blocks associated with these channels. Specifically, the OPE of any two ϵ fields yields only the identity operator ($\epsilon \times \epsilon = \mathbf{1}$), which simplifies the computation as $\langle \epsilon\epsilon\epsilon\sigma\sigma \rangle$ can be related to lower-point functions (e.g., $\langle \epsilon\sigma\sigma \rangle$) via the identity channel, plus contributions from the $\sigma \times \sigma$ channel.

3. Structure of the Correlation Function

The 5-point correlation function for primary fields ϕ_i at positions $z_i, \bar{z}_i$ takes the form:

$$\langle \phi_1(z_1, \bar{z}_1) \dots \phi_5(z_5, \bar{z}_5) \rangle = \left(\prod_{1 \leq i < j \leq 5} |z_i - z_j|^{-2\Delta_{ij}} \right) \times F(x_{ij}, \bar{x}_{ij})$$

Where the prefactor accounts for the global conformal symmetry, and F is a function of the independent cross-ratios. For $N = 5$, there are two independent complex cross-ratios.

For the specific operator content $\langle \epsilon_1 \epsilon_2 \epsilon_3 \sigma_4 \sigma_5 \rangle$, the kinematic prefactor is determined by:

- $\Delta_1 = \Delta_2 = \Delta_3 = 1/2$
- $\Delta_4 = \Delta_5 = 1/16$

The function can be expanded in terms of 5-point conformal blocks. Due to the fusion rule $\epsilon \times \epsilon = \mathbf{1}$, the leading contribution in the channel where ϵ_1 and ϵ_2 fuse is proportional to the 3-point function $\langle \epsilon_3 \sigma_4 \sigma_5 \rangle$.

$$\langle \epsilon(z_1) \epsilon(z_2) \epsilon(z_3) \sigma(z_4) \sigma(z_5) \rangle \sim \frac{1}{|z_1 - z_2|^2} \langle \epsilon(z_3) \sigma(z_4) \sigma(z_5) \rangle + \dots$$

4. Input Data for Computation

The specific coordinates required for the calculation are:

Case (1): Complex coordinates

- $x_1 = 1 + i$
- $x_2 = 2$
- $x_3 = 3$
- $x_4 = 4$
- $x_5 = 5$

Case (2): Real coordinates

- $x_1 = 1$
- $x_2 = 2$
- $x_3 = 3$
- $x_4 = 4$
- $x_5 = 5$

References

[1] P. Di Francesco, P. Mathieu, and D. Sénéchal, *Conformal Field Theory*, Springer, 1997.